

Optical Thermometry in Monolayer WS₂: Raman and Photoluminescence Spectroscopy from 5 K to 1273 K

G. Lee¹, A. Borel¹, T. Taniguchi², K. Watanabe³, F. Sirotti¹, and F. Cadiz¹

¹Laboratoire de Physique de la Matière Condensée, CNRS, Ecole polytechnique, Institut Polytechnique de Paris, 91120 Palaiseau, France

²International Center for Materials Nanoarchitectonics, National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, Japan

³Research Center for Functional Materials, National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, Japan

*Email: fabian.cadiz@polytechnique.edu

Abstract

Reliable temperature measurement is essential for characterizing and engineering two-dimensional (2D) semiconductor devices, yet optical thermometry of 2D materials has so far been demonstrated only over narrow temperature windows spanning a few hundred kelvin. Here, we establish a comprehensive optical thermometry framework for monolayer WS₂ from 5 K to 1273 K. Using two independent optical probes on complementary samples, we show that: (i) the out-of-plane A_{1g} Raman mode shifts linearly with temperature above 100 K, with non-encapsulated WS₂ exhibiting a slope of $-0.0138 \text{ cm}^{-1}/\text{K}$ (fitted over 298–1223 K) and hBN-encapsulated samples a measurably smaller slope of $-0.0121 \text{ cm}^{-1}/\text{K}$ (fitted over 100–1273 K); (ii) photoluminescence spectra measured over the same 5–1273 K range provide an independent optical indicator: the neutral-exciton emission X_A^0 redshifts continuously with temperature and is well described by the semi-empirical Pässler model. The two indicators probe distinct physical mechanisms, lattice anharmonicity and bandgap renormalization, and together provide contact-free thermometry for monolayer WS₂ from 5 K to 1273 K, an approach applicable to 2D semiconductors more broadly.

Introduction

The isolation of two-dimensional (2D) atomic crystals, beginning with graphene and extending to a broad family of layered compounds [1], has established a class of materials whose properties are set by their reduced dimensionality. Among them, monolayer transition metal dichalcogenides (TMDs) are direct-gap semiconductors whose optical response is dominated by tightly bound excitons even at room temperature [2, 3]. The strong Coulomb interaction and reduced dielectric screening in these atomically-thin materials give exciton binding energies of several hundred meV and make the optical response acutely sensitive to the dielectric surroundings [4, 5].

High-temperature annealing is emerging as a route to defect engineering in 2D semiconductors: in monolayer MoS₂ it generates chalcogen-vacancy defects with characteristic optical signatures [6], and in hBN-encapsulated monolayer WS₂ annealing near 1100 K produces ultra-narrow emission lines that act as single-photon emitters [7]. Controlling such processes requires reliable knowledge of the temperature of the 2D layer itself, which cannot be obtained by attaching a contact sensor to a layer one atom thick. The optical response of the monolayer offers a way out: both its excitonic emission and its Raman spectrum shift measurably with temperature, so the spectrum itself can serve as a built-in, contact-free thermometer.

Optical thermometry based on photoluminescence (PL) and Raman spectroscopy is well established as a non-contact probe at the nanoscale [8], and has been extended to 2D materials such as graphene, where

the G and 2D Raman modes provide local temperature readout [9]. In the case of monolayer MoS₂ and WS₂, temperature-dependent Raman spectroscopy has been used to extract thermal conductivities [10, 11], with linear peak shifts reported over limited temperature windows, typically spanning a few hundred kelvin below ~ 550 K [12, 13]. However, the combination of (i) an ultra-wide temperature range from cryogenic to near-dissociation temperatures, (ii) simultaneous access to both Raman and PL observables, and (iii) a quantitative assessment of the robustness of these indicators across different encapsulation conditions has not been demonstrated to date.

A key practical challenge is that high-power laser illumination, required for sufficient Raman signal in monolayer TMDs, induces strong photodoping in non-encapsulated samples, as established for the related molybdenum dichalcogenides [14], making PL inaccessible after Raman acquisition. Encapsulation of the TMD monolayer between hexagonal boron nitride (hBN) layers suppresses this photodoping effect [14] and dramatically narrows excitonic linewidths [15, 16], enabling PL spectroscopy under the same conditions as Raman. The trade-off is a reduction in the Raman signal due to the additional hBN layers, leading to missing data points at certain temperatures. The two measurement configurations are therefore complementary rather than interchangeable.

In this work, we exploit this complementarity to establish an optical thermometry framework for monolayer WS₂ spanning 5 K to 1273 K—a range that, to our knowledge, has not been covered by a single 2D material system in a single study. Both samples are mounted on a suspended SiC micro-heater membrane inside a cryostat, so that the monolayer can be heated locally while the surrounding environment stays cold. The refractory character of SiC allows the membrane to be operated repeatedly up to 1273 K, so that the same sample can be measured over the full range without changing the optical setup. The non-encapsulated sample gives the Raman thermal response from room temperature to 1223 K, while the hBN-encapsulated sample provides PL over the full range together with complementary Raman data. Calibrating both signatures against a known temperature turns the spectrum itself into the thermometer: a single Raman or PL acquisition then reports the temperature of the monolayer at that spot.

Experimental Methods

Sample Preparation

For the non-encapsulated sample (Experiment 1), monolayer WS₂ was mechanically exfoliated from a bulk crystal and transferred directly onto the suspended SiC micro-membrane of a commercial micro-heater chip (EDEN Instruments). For the encapsulated sample (Experiment 2), hBN/WS₂/hBN heterostructures were assembled by dry-stamp transfer [17] and subsequently placed onto a second membrane chip. The SiC membrane is 120 nm thick, suspended between doped Si contacts, and enables uniform Joule heating of the central region up to ~ 1200 K when a dc current is applied. The temperature of the membrane was calibrated against the applied electrical current by the manufacturer [7].

Temperature Control

Two complementary temperature stages were used for the encapsulated sample. From 5 K to room temperature, the sample was cooled in a He-flow cryostat and heated uniformly using the cryostat’s built-in heater. In this regime, the cryostat thermometer measures the temperature of the entire cold finger, which is in equilibrium with the sample. From room temperature to ~ 1273 K, the SiC membrane was heated resistively under vacuum ($\sim 10^{-3}$ mbar) while the cryostat environment remained at low temperature. The current-to-temperature conversion was obtained from the manufacturer’s calibration. For the non-encapsulated sample, only high-temperature measurements were performed (room temperature to 1273 K under vacuum).

Optical Measurements

All measurements used a 514.5 nm continuous-wave laser (Cobolt). For Raman spectroscopy, the excitation power was ~ 1 mW; for PL spectroscopy of the encapsulated sample, the power was reduced to $\lesssim 10$ μ W to avoid photo-induced heating. The laser was focused to a ~ 1 μ m spot through a 50 $\times$ long-working-distance objective. The scattered or emitted light was collected by the same objective, filtered with volume Bragg

notch filters centered at 514.5 nm (for Raman), and dispersed by a grating spectrometer onto a Peltier-cooled CCD. For each temperature step, the current was increased gradually, held at the target value for 30 min, and then reduced slowly back to zero before the next acquisition. PL spectra were fitted with a pseudo-Voigt profile (see Supporting Information) to extract the peak energy E_0 and the total full-width at half-maximum (FWHM) at each temperature.

Results

Raman Spectroscopy as a Thermal Indicator

Non-encapsulated WS₂ (RT to 1223 K)

A representative Raman spectrum of the non-encapsulated WS₂ monolayer at room temperature is shown in Fig. 1(c). In addition to the principal zone-center modes E_{2g}¹(Γ) (~352 cm⁻¹) and A_{1g}(Γ) (~414 cm⁻¹), several zone-boundary combination modes are resolved, including 2LA(M) (~349 cm⁻¹), 3LA(M) (~517 cm⁻¹), and A_{1g}(Γ)+LA(M) (~576 cm⁻¹), consistent with prior reports on monolayer WS₂ [18]. The A_{1g}(Γ) mode is used as the primary thermal indicator in what follows; the overlapping E_{2g}¹(Γ)+2LA(M) doublet is analysed separately in the Supporting Information (Sec. S1), where it serves as a secondary, qualitative check.

Figure 2(a) shows a two-dimensional intensity map of the Raman spectra as a function of temperature from room temperature to 1223 K. With increasing temperature, both peaks redshift continuously and broaden, while their intensity decreases. The redshift arises from two contributions: thermal expansion of the lattice, which lowers the phonon frequency through the Grüneisen relation, and anharmonic phonon-phonon coupling, whose three- and four-phonon decay channels become increasingly populated [12]. The signal vanishes above ~1223 K, consistent with thermal dissociation of the WS₂ lattice through sulfur sublimation [19] (see also Supporting Information).

The A_{1g} peak position extracted from these spectra is shown in Fig. 2(c) (red squares). A linear fit restricted to $T \geq 100$ K yields:

$$\omega_{A_{1g}}(T) = \omega_0 + \left(\frac{d\omega}{dT} \right) T, \quad (1)$$

with a slope $d\omega/dT = -0.0138 \pm 0.0002$ cm⁻¹/K and intercept $\omega_0 = 417.95$ cm⁻¹ ($R^2 = 0.995$, $T = 298$ –1223 K); the same dataset analysed with the anti-Stokes A_{1g} line included gives -0.0134 cm⁻¹/K [7]. A linear fit to the A_{1g} shift of monolayer WS₂ on quartz by Huang *et al.* gave -0.0122 cm⁻¹/K over 84–543 K, although those authors restricted their quantitative analysis of this mode to below 380 K, above which strain from wrinkling of the monolayer on the substrate distorted the response [12]. Reported first-order coefficients for the A_{1g} mode span -0.006 to -0.0149 cm⁻¹/K across studies [12], so that values obtained on different samples, substrates and fitting ranges are not directly comparable; the two configurations measured here were acquired on the same platform with identical excitation and fitting procedures, making the difference between them, quantified below, a controlled comparison. The A_{1g} mode is preferred as the primary thermometric indicator because, unlike the overlapping E_{2g}¹(Γ)+2LA(M) doublet, it is spectrally isolated, so that its position can be determined unambiguously at every temperature.

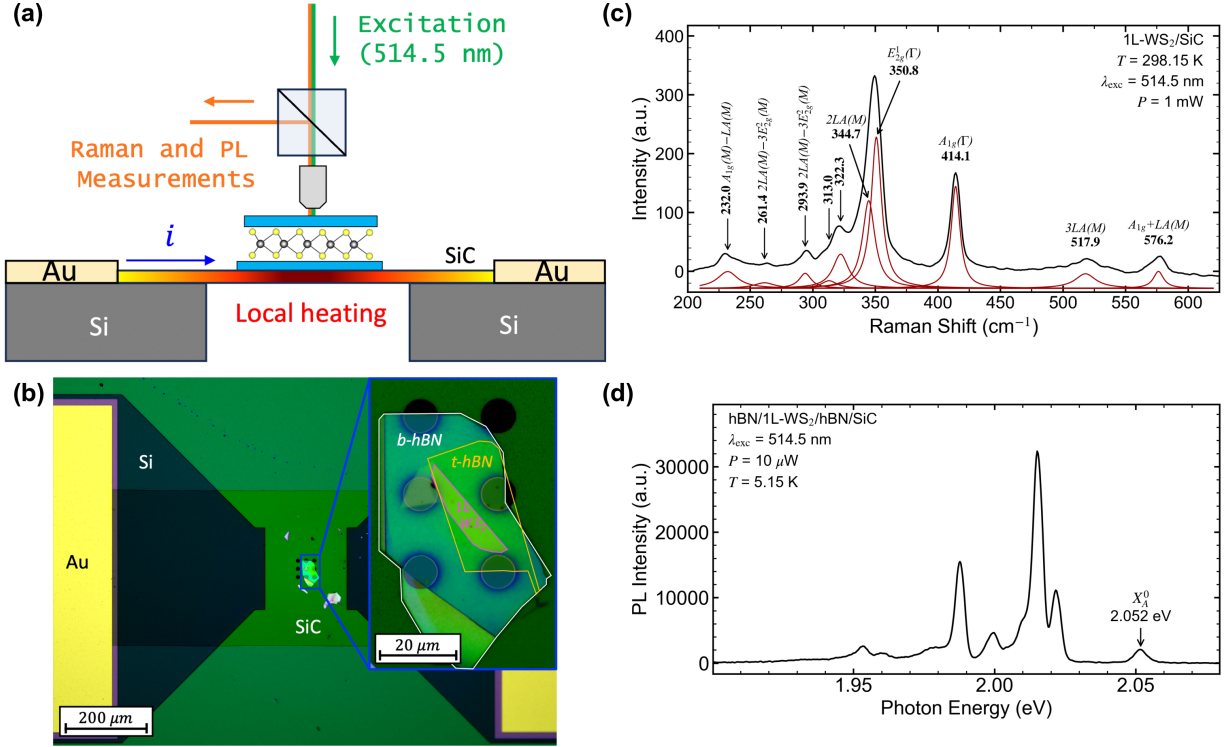


Figure 1: **Experimental platform and reference spectra.** (a) Schematic of the in-situ measurement platform: hBN/WS₂/hBN heterostructure transferred onto the suspended SiC micro-membrane of the micro-heater chip, mounted inside a cryostat. (b) Optical image of the encapsulated WS₂ flake on the SiC membrane under white-light illumination. (c) Room-temperature Raman spectrum of non-encapsulated monolayer WS₂ ($\lambda_{\text{exc}} = 514.5$ nm, $P = 1$ mW), including Lorentzian peak fits for the $E_{2g}^1(\Gamma)$, $2LA(M)$, $A_{1g}(\Gamma)$, and higher-order combination modes. High-power illumination induces photodoping in the absence of hBN encapsulation, precluding PL measurements on this sample. (d) PL spectrum of hBN-encapsulated WS₂ at 5 K, showing the neutral exciton X_A^0 , trion, and dark-exciton-related features ($\lambda_{\text{ex}} = 514.5$ nm, $P = 10$ μ W).

hBN-encapsulated WS₂ (5 K to 1273 K)

The hBN-encapsulated sample allows simultaneous Raman and PL measurements under identical laser conditions, since hBN encapsulation suppresses the photodoping that precludes PL in the non-encapsulated sample [14]. The Raman and PL data presented in this and the following section were acquired on the same encapsulated sample.

Figure 2(b) shows the two-dimensional Raman intensity map of the encapsulated sample over the full temperature range from 5 K to 1273 K. The overall signal level is reduced compared to the non-encap sample due to partial attenuation by the hBN layers. The map reveals a temperature-dependent intensity redistribution between the two modes: at low temperature (below ~ 100 K), the $E_{2g}^1 + 2LA$ mode dominates; at high temperature, the A_{1g} mode becomes the stronger feature; at intermediate temperatures both modes are weak, leading to gaps in the extracted peak positions. One factor that plausibly contributes to the redistribution is a change in resonance condition: at 514.5 nm, monolayer WS₂ sits in a second-order Raman resonance that makes the $2LA(M)$ mode roughly twice as intense as the first-order $A_{1g}(\Gamma)$ mode, an enhancement specific to this excitation wavelength and governed by the electronic band structure [18]. Because the excitation energy is fixed while the electronic transitions of the sample redshift by ~ 0.4 eV over the range studied (Fig. 3(b)), the detuning between the two changes continuously with temperature, and the second-order mode is expected to be far more sensitive to this than the first-order one. This alone does not account for the observation that *both* modes weaken at intermediate temperatures, and the physical origin of the redistribution is under investigation and will be addressed in a forthcoming study.

Despite the missing data points at intermediate temperatures, Fig. 2(c) (blue circles) shows that the A_{1g} peak positions obtained from the encapsulated sample across the full accessible range (5 K to RT from cryogenic measurements; RT to 1273 K from micro-heater measurements) follow the same qualitative linear trend established for the non-encapsulated sample, indicating that the underlying phonon-softening mechanism is intrinsic to the WS_2 lattice regardless of encapsulation, although the two samples differ measurably in thermal sensitivity, as quantified below. A linear fit restricted to $T \geq 100$ K yields a slope $d\omega/dT = -0.0121 \pm 0.0001 \text{ cm}^{-1}/\text{K}$ and intercept $\omega_0 = 419.55 \text{ cm}^{-1}$ ($R^2 = 0.997$), a magnitude $\sim 12\%$ smaller than in the non-encapsulated sample. Because this fit pools cryostat-calibrated points below room temperature with micro-heater-calibrated points above it, whereas the non-encapsulated fit uses micro-heater calibration alone, we checked that the difference is not an artefact of combining the two temperature scales: restricting the encapsulated fit to $T \geq 298$ K, so that it too rests on the micro-heater calibration only, gives $-0.0121 \text{ cm}^{-1}/\text{K}$, unchanged within the fit uncertainty. This difference is in the direction expected if the hBN layers mechanically constrain the monolayer: the in-plane thermal expansion coefficient of hBN is negative [20], whereas that of WS_2 is positive [12], so an encapsulated monolayer cannot expand as freely as a bare one and the thermal-expansion contribution to the phonon softening is correspondingly reduced. Testing this quantitatively would require an independent measurement of the residual strain and is left for future work.

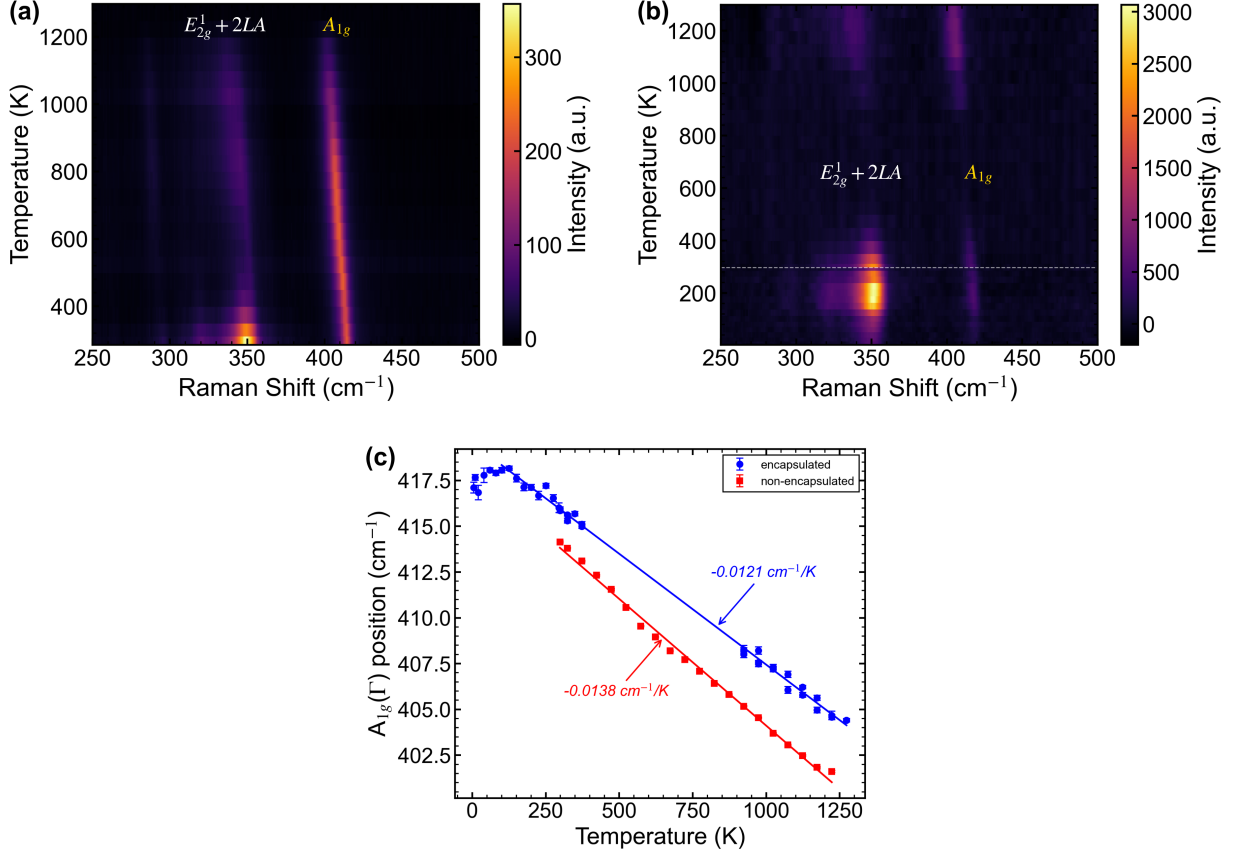


Figure 2: **Raman thermal indicator: non-encapsulated and hBN-encapsulated WS₂.** (a) Two-dimensional map of Raman intensity versus temperature (298–1223 K) and Raman shift for the non-encapsulated sample. Both the $E_{2g}^1 + 2LA$ and A_{1g} modes redshift continuously with increasing temperature. (b) Two-dimensional map of Raman intensity versus temperature (5–1273 K) and Raman shift for the hBN-encapsulated sample. The white dashed line marks the boundary between the cryostat (5 K–RT) and micro-heater (RT–1273 K) acquisition stages. A temperature-dependent intensity redistribution between the $E_{2g}^1 + 2LA$ (dominant at low T) and A_{1g} (dominant at high T) modes is observed, with both peaks weak at intermediate temperatures. (c) $A_{1g}(\Gamma)$ peak position versus temperature extracted from both samples: non-encapsulated (red squares) and hBN-encapsulated (blue circles). Solid lines are linear fits restricted to $T \geq 100$ K, with slopes of -0.0138 cm⁻¹/K (non-encapsulated, $R^2 = 0.995$) and -0.0121 cm⁻¹/K (hBN-encapsulated, $R^2 = 0.997$). Raman and PL measurements on the encapsulated sample were performed simultaneously (see Fig. 3).

Photoluminescence as a Thermal Indicator

Spectral evolution and peak energy. Figure 3(a) shows the evolution of the PL spectrum of the hBN-encapsulated WS₂ monolayer, plotted as a waterfall of six representative spectra spanning 5 K to 1023 K. At 5 K the emission is dominated by lower-lying excitonic complexes, and the neutral exciton X_A^0 at $E_0 = 2.052$ eV, with a narrow Voigt linewidth of ~ 5.3 meV, is one of the weakest features of the spectrum (Fig. 1(d)). This reflects the darkish nature of tungsten-based monolayers, in which the excitonic ground state is optically inactive, so that photocarriers relax preferentially into dark and charged states [21, 22]. With increasing temperature, thermal activation redistributes population into the bright X_A^0 state, whose integrated intensity rises by two orders of magnitude up to ~ 175 K (Supporting Information, Fig. S5); at the same time the lower-lying complexes broaden and merge, and above ~ 300 K the spectrum consists of a single X_A^0 line. On further heating this line redshifts monotonically and broadens, while its emission efficiency falls

through thermally activated non-radiative recombination [23] and the escape of excitons from the radiative window [24].

The neutral-exciton peak energy $E_0(T)$ extracted from Voigt fits is shown in Fig. 3(b). The redshift reflects the interplay of two mechanisms: thermal lattice expansion, which reduces the orbital overlap and lowers the bandgap, and electron-phonon coupling, which renormalizes the quasiparticle energies [25]. The data are excellently described by the Pässler model [25, 26]:

$$E_0(T) = E_0(0) - \frac{\alpha\Theta}{2} \left[\sqrt[p]{1 + \left(\frac{2T}{\Theta}\right)^p} - 1 \right], \quad (2)$$

where $E_0(0)$ is the zero-temperature bandgap, α is the high-temperature slope, Θ is the mean phonon temperature, and p is a dimensionless dispersion exponent. The best-fit parameters obtained here are: $E_0(0) = 2.0491 \pm 0.0036$ eV, $\alpha = (3.741 \pm 0.069) \times 10^{-4}$ eV/K, $\Theta = 280.1 \pm 28.8$ K, $p = 4.04 \pm 2.23$ (the large relative uncertainty on p reflects its known correlation with Θ in this model) ($R^2 = 0.9964$, RMSE = 7.9 meV; see Table S1). These are consistent with previously reported values for encapsulated WS₂ [7], with small deviations attributed to differences in local strain and dielectric environment between samples. A comparison of the Pässler model with the Varshni and O'Donnell-Chen models is provided in the Supporting Information; both the Pässler and O'Donnell-Chen models reproduce the data equivalently well, while the Varshni model underestimates the curvature at low temperature.

The inversion of Eq. (2) allows the temperature to be read directly from the measured E_0 , providing a model-based, contact-free thermometer calibrated against the cryostat (5 K to RT) and the manufacturer's chip calibration (RT to 1273 K). The practical lower limit of this PL thermometer is set by the spectral resolution of the detection system. Towards higher temperature, the X_A^0 peak broadens substantially (from ~ 5.3 meV at 5 K to ~ 140 meV at 1273 K) and becomes progressively less well defined, so that its position is determined with decreasing precision; the measurements reported here extend to 1273 K. The root-mean-square deviation of the Pässler fit residuals is 7.9 meV, corresponding to a temperature uncertainty of ~ 21 K in the high-temperature (linear) regime, estimated as RMSE/α ; the uncertainty is larger at cryogenic temperature, where $dE_0/dT \rightarrow 0$.

Spectral linewidth. The X_A^0 emission broadens monotonically with temperature, the total Voigt FWHM increasing from ~ 5.3 meV at 5 K to ~ 140 meV at 1273 K (Supporting Information, Fig. S5). The narrow low-temperature linewidth reflects the quality of the hBN-encapsulated heterostructure [15, 16], while the progressive broadening at higher temperature is consistent with an increasing phonon-scattering-limited dephasing rate [27, 28]. The linewidth is not used as a thermometric indicator here: it is strongly sample-dependent, and at the highest temperatures the peak extends beyond the low-energy edge of the fitting window, so that the width is no longer reliably determined.

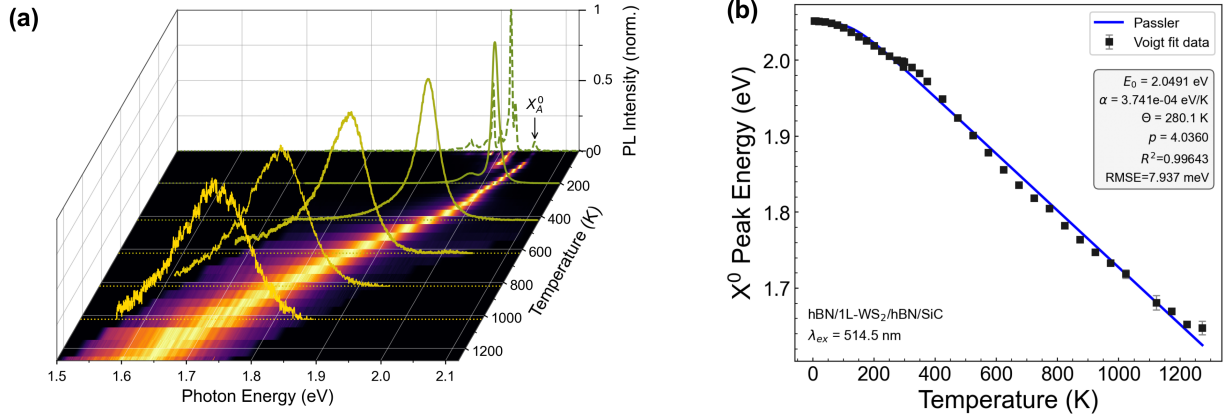


Figure 3: **Photoluminescence thermal indicator in hBN-encapsulated monolayer WS₂.** (a) Waterfall plot of representative PL spectra (5 K to 1023 K) as a function of photon energy, normalized to the peak intensity at each temperature, overlaid on the full temperature-dependent intensity map. The X_A^0 neutral-exciton peak, visible at low temperature, redshifts and broadens continuously with increasing temperature. (b) Neutral-exciton peak energy E_0 extracted from Voigt fits (black squares, with error bars) and the Pässler model fit (blue line), Eq. (2), with best-fit parameters and fit quality annotated.

Discussion

Our results establish monolayer WS₂ as a broadband optical thermometer whose two independent indicators—Raman peak position and PL peak energy—are applicable over different, overlapping temperature windows that together span 5 K to 1273 K.

The Raman indicator is the simpler of the two: a single linear calibration curve suffices above 100 K. The hBN-encapsulated sample, however, shows a measurably different calibration from the bare monolayer, its slope being 12% smaller (-0.0121 vs. -0.0138 cm^{-1}/K). Because the slope is a differential quantity, this difference is independent of the absolute wavenumber calibration, and it sets a floor on the accuracy of any transferred calibration: using one sample’s slope to read the other’s temperature leaves a systematic error of 12%, or ~ 120 K at 1000 K. A Raman calibration obtained on a bare monolayer therefore cannot be carried over to an encapsulated heterostructure without re-calibration. What this work establishes is the range over which either calibration remains linear, from 100 K to 1273 K, substantially broader than the window previously reported for WS₂ Raman thermometry [12].

The PL thermometer relies on the neutral-exciton peak energy, for which the Pässler model provides a physically grounded, analytic expression that can be inverted to yield temperature from a single spectral acquisition. Importantly, the PL thermometer extends the range to cryogenic temperatures (5 K), which the Raman thermometer cannot easily access due to the difficulty of distinguishing small thermal shifts in the nearly temperature-independent phonon spectrum at low T .

The two indicators are complementary in a specific sense. Below ~ 100 K the Raman shift is too weakly temperature-dependent to resolve, whereas the PL peak energy still tracks temperature with sub-kelvin precision; PL is therefore the only usable indicator in the cryogenic regime. At high temperature the two are comparable: converting the peak-position uncertainty of each fitted spectrum into a temperature through the corresponding sensitivity (σ_{E_0}/α for PL, $\sigma_\omega/|d\omega/dT|$ for Raman) gives, averaged over all points above 900 K, ~ 12 K for PL and ~ 13 K for Raman. The practical distinction there is not precision but sample configuration, since photodoping precludes PL on non-encapsulated samples [14], leaving the Raman shift as the only available indicator for bare monolayers.

Finally, it is useful to place this range in the context of Raman thermometry on other 2D materials, where the published temperature windows are considerably narrower: 100–320 K for suspended monolayer MoS₂ [10], 303–448 K for monolayer MoS₂ on SiO₂ [13], and 84–543 K for monolayer WS₂ on quartz [12], the last restricted to below 380 K for the A_{1g} mode by substrate-induced wrinkling. The linear behaviour

reported here therefore extends over a range several times wider than in any of these studies. Departures from linearity are present but small: including a quadratic term changes the fitted frequency by at most 0.4 cm^{-1} out of a total excursion of 12.5 cm^{-1} , so a single linear calibration is retained throughout.

The two samples reach their upper limits for different reasons. The non-encapsulated monolayer dissociates through sulfur sublimation [19]: its Raman signal is lost above $\sim 1223 \text{ K}$ and the flake is no longer visible afterwards (Supporting Information, Sec. S6). The encapsulated monolayer survives 1273 K intact, with both Raman and PL still measurable, and the measurement was stopped there to protect the heater chip rather than because the sample failed. hBN encapsulation therefore extends the thermal stability of the monolayer as well as its spectral quality.

Two features of the platform make the extended range accessible: the suspended SiC membrane allows the monolayer to be heated locally while the surrounding cryostat stays cold, and hBN encapsulation sharpens the excitonic linewidth sufficiently for reliable peak fitting down to 5 K [15, 16]. The large exciton binding energy of monolayer WS_2 ($\sim 300 \text{ meV}$) [5] ensures a well-defined excitonic peak at all temperatures studied, and the use of a single excitation wavelength (514.5 nm) for both Raman and PL simplifies the measurement.

Conclusion

We have established an optical thermometry framework for monolayer WS_2 spanning 5 K to 1273 K using two independent optical indicators. The A_{1g} Raman mode shifts linearly with temperature above 100 K , with slopes of $-0.0138 \text{ cm}^{-1}/\text{K}$ for the bare monolayer ($298\text{--}1223 \text{ K}$) and $-0.0121 \text{ cm}^{-1}/\text{K}$ for the hBN-encapsulated heterostructure ($100\text{--}1273 \text{ K}$); the 12% difference between them means that a calibration obtained on one configuration cannot be transferred to the other. The neutral-exciton PL peak energy, described by the Pässler model, provides a second indicator over the same range, with a fit residual corresponding to $\sim 21 \text{ K}$ in the high-temperature regime.

Two questions remain open. The origin of the temperature-dependent intensity redistribution between the $E_{2g}^1+2\text{LA}$ doublet and the A_{1g} mode, which removes both features at intermediate temperatures in the encapsulated sample, is still under investigation. And the upper limit reached here was set by the heater chip rather than by the encapsulated monolayer, which survived 1273 K intact; how much further hBN encapsulation extends the usable range remains to be established.

Acknowledgements

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Data Availability

The data that support the findings of this study will be made available in Zenodo upon publication.

Supporting Information

The Supporting Information contains: (S1) Individual versus merged-peak fitting of the $E_{2g}^1+2\text{LA}$ Raman doublet for both samples; (S2) Temperature-dependent Raman spectra of the hBN-encapsulated sample; (S3) Bandgap model comparison (Varshni, Pässler, O'Donnell–Chen) with fit residuals; (S4) Voigt fitting procedure and representative fit quality; (S5) Temperature dependence of the PL spectral linewidth and integrated X_A^0 intensity; (S6) WS_2 monolayer after thermal dissociation.

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